

Session 2: How Software Works Define software application and its components. Explain software architecture layers: UI? Business Logic? Database. Compare desktop, web, and mobile apps. Explain client-server communication with example (browser request? backend? database? response). Trainer Demo: Use Draw.io or Lucid chart AI to visually show data flow (e.g., for "Instagram feed refresh").

TASKS:

1. What are the three main layers of software architecture? Write one line about each.

1. Presentation Layer (UI Layer)
2. Business Logic Layer (Application Layer)
3. Data Layer (Database Layer)

1. Presentation Layer (UI Layer) – Handles user interaction and displays information to users.
2. Business Logic Layer (Application Layer) – Processes data and implements the application's rules and functionality.
3. Data Layer (Database Layer) – Stores, retrieves, and manages data in the database.

2. Define client and server in your own words with one real-life example.

Client: A client is a device or application that requests services or data from another computer.

Server: A server is a computer or program that provides services or data to clients.

Real-life Example: When you open the website of Google in your browser, your browser acts as the **client**, and Google's web server acts as the **server** by sending the requested web pages to your device.

3. Write 2 differences each between desktop apps, web apps, and mobile apps.

Desktop Apps:

- Desktop apps are installed directly on a computer.
- They can often work without an internet connection.
- Usually designed for a specific operating system (Windows, macOS, etc.).

Web Apps:

- Web apps run inside a web browser.
- They usually require an internet connection to work.

Mobile Apps

- Mobile apps are designed for smartphones and tablets.
- They can use mobile device features like the camera, GPS, and touch screen.

4. Name any 3 databases and mention whether each is SQL or NoSQL.

MySQL – SQL Database (Relational)

MongoDB – NoSQL Database (Document-based)

Oracle Database – SQL Database